Benjamin Chou

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Professional Summary

Final-year Purdue ECE PhD student researching audio, music, and multimodal AI/ML: personalized speech enhancement, note-level music separation, music error detection with Transformers, and training-free Multimodal LLMs. Proficient in Audio DSP, Python, PyTorch, TensorFlow, AWS (SageMaker), and HPC. Organized three AI4Music workshops.

Education

Ph.D. in Electrical & Computer Engineering | *Purdue University*, West Lafayette, IN — 2023 - Present
- GPA: 3.8/4.0
- Expected graduation: Sep 2027 (tentative)

B.Sc. in Electrical Engineering | *National Cheng Kung University*, Tainan, Taiwan — 2020 - 2023
- GPA: 4.0/4.3

Experience

Returning Intern | *Google X*, Mountain View, CA — Jan 2027 - Jul 2027
- Returning to Google X to continue work on LLM graph orchestration, loop engineering, and code-review agents.

Audio Applied Research Scientist Intern | *Shure*, Niles, IL — Jul 2026 - Present
- Researching personalized speech enhancement and target speech extraction for multi-speaker conferencing systems.
- Investigating small, real-time-capable generative models for conferencing audio (**flow matching**, **MeanFlow**).

PhD Resident | *Google X*, Mountain View, CA — Apr 2026 - Jul 2026
- Automated agent development with LLM graph orchestration and loop engineering, producing prototypes with novel capabilities that matched or exceeded those of human-designed pipelines.
- Built an LLM code-review agent that replaced the team's internal agent with stronger review capabilities (**LangChain**, **Vertex AI**, **RAG**, **agent skills**, **Google Cloud Platform** (**GCP**), **GitHub CLI**, **Linear**).

Applied Research Scientist Intern | *Reality Defender*, Manhattan, New York — May 2025 - Sep 2025
- Developed in-context learning with large audio language models for SOTA in-the-wild speech deepfake detection (+54.9% avg Macro F1 improvement) (**NLP**, **RAG**, **ICL**, **Audio LLMs**, **model routing**).

Graduate Research Assistant | *Purdue University (Dr. Yung-Hsiang Lu)*, West Lafayette, IN — Jan 2024 - Present
- Developed music tutoring models: 6× higher F1 (AAAI 2025) and 2× better missed-note detection with LadderSym (ICLR 2026) (**PyTorch**, **NLP**, **Audio Transformers**).
- Built generative music-mistake pipelines on HPC and Linux, producing two datasets.
- Organized 'AI for Music' workshops at AAAI 2025, IEEE ICME 2025, and NeurIPS 2025 (**leadership**, **communication**).

Research Assistant | *National Cheng Kung University (Dr. Hsun-Ping Hsieh)*, Taiwan — Mar 2023 - Aug 2023
- Raised cancer-screening invitation accuracy to 83% with scikit-learn tree classifiers (**ML**, **statistics**, **data science**).
- Improved flash-chip yield by 17% with ensembling and Gaussian processes (**NumPy**, **pandas**).

Projects

Code Understanding with GitHub Issues and RAG | (**ChatGPT API**, **REST**, **Git**)
- Built a RAG-based doc tool with GitHub issues; raised README clarity by 116% and code clarity by 16%.

Multi-Agent Atari Self-Play | (**OpenAI Gym**, **PettingZoo**, **PyTorch**, **RL**, **Multi-agent systems**)
- Used DQN/PPO self-play to improve Atari win rates by 11–32%.

Spectral Image Inpainting | (**MATLAB**, **C++**, **CUDA**)
- Recovered 98% corrupted hyperspectral data at 0.02 RMSE using convex optimization and deep learning.

Honors & Awards

- Outstanding Student Scholarship: top 2 students in the NCKU-Purdue program (2020 - 2023).
- AAAI Student Scholarship (2025), selected for NeurIPS volunteer (2025).

Publications

- **B. Chou**, P. Jajal, N. J. Eliopoulos, J. C. Davis, G. K. Thiruvathukal, K. Y.-J. Yun, H.-W. Dong, Y.-H. Lu, "On a Separate Note: Robust Score-Informed Note Separation with a Two-Stream TFC-TDF U-Net and Adaptive Set Ownership", *Submitted to ICASSP 2027*
- **B. Chou**, Y. Zhu, S. Koppisetti, "ICLAD: In-Context Learning with Comparison-Guidance for Audio Deepfake Detection", *ACL 2026 Findings*
- P. Jajal, N. J. Eliopoulos, **B. Chou**, G. K. Thiravathukal, J. C. Davis, Y. Lu, "AdaPerceiver: Transformers with Adaptive Width, Depth, and Tokens", *CVPR 2026 Findings*
- P. Jajal, N. J. Eliopoulos, **B. Chou**, G. K. Thiravathukal, J. C. Davis, Y. Lu, "Inference-Time Alignment of Diffusion Models with Evolutionary Algorithms", *CVPR 2026 Findings*
- **B. Chou**, P. Jajal, N. J. Eliopoulos, J. C. Davis, G. K. Thiravathukal, K. Y.-J. Yun, Y.-H. Lu, "LadderSym: A Multimodal Interleaved Transformer for Music Practice Error Detection", *ICLR 2026 (top 3.85%)*
- **B. Chou**, P. Jajal, N. J. Eliopoulos, T. Nadolsky, C.-Y. Yang, N. Ravi, J. C. Davis, K. Y.-J. Yun, Y. Lu, "Detecting Music Performance Errors with Transformers", *AAAI 2025*
- P. Jajal, N. J. Eliopoulos, **B. Chou**, G. K. Thiravathukal, J. C. Davis, Y.-H. Lu, "Token Turing Machines are Efficient Vision Models", *WACV 2025*

Patent Pending

- System and Method for Detecting Musical Performance Errors, Patent Application 2025.
- ICLAD: In-Context Learning with Comparison-Guidance for Audio Deepfake Detection, Provisional Patent Application 2025.